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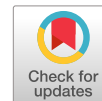
Systematic investigation of charge transport and thermoelectric properties in semicrystalline polymers: electrochemical doping effects on doping level and temperature dependence using one sample

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Systematic investigation of charge transport and thermoelectric properties in semicrystalline polymers: electrochemical doping effects on doping level and temperature dependence using one sample

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We developed a system to measure the temperature dependence of the Seebeck coefficient (S), electrical conductivity, and magnetoconductivity of conducting polymers with controlled doping levels, using an electrolyte-gated transistor structure on a single sample. In the semicrystalline polymer poly[2,5-bis(3-alkylthiophen-2-yl)thieno[3,2-b]thiophene] (PBTTT), S reflects hopping-like transport in the low-doped state, while metallic behavior dominates in the highly-doped state. At low temperatures in the highly-doped regime, S deviates from metallic behavior, showing an unusual trend attributed to the coexistence of localized and delocalized transport processes. Magnetoconductance measurements down to 80 mK confirm this coexistence, highlighting the interplay between two-dimensional weak localization and variable-range hopping. These findings reveal the complexity of transport mechanisms in conducting polymers and provide essential insights for optimizing thermoelectric and electronic devices through careful control of doping and transport properties. © 2025 The Author(s). Published on behalf of The Japan Society of Applied Physics by IOP Publishing Ltd

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Conducting polymers, which offer unique advantages such as high flexibility, lightweight nature, and ease of processing, are considered highly promising materials for various applications, including thermoelectric generators.^{1,2)} However, the charge transport and thermoelectric mechanisms in doped conducting polymers remain unclear. For instance, an unusual power-law relationship between the Seebeck coefficient (S) and electrical conductivity (σ) has been reported at room temperature.^{3–7)} Additionally, the temperature dependence of the Seebeck coefficient, $S(T)$ and the electrical conductivity, $\sigma(T)$ exhibit distinct behaviors. Specifically, in previous studies on chemically doped semicrystalline polymer poly[2,5-bis(3-alkylthiophen-2-yl)thieno[3,2-b]thiophene] (PBTTT), a linear $S(T)$ has been observed, suggesting a metallic and delocalized state. On the other hand, in the same sample, $\sigma(T)$ indicates localized hopping transport.^{8–10)}

These findings suggest that the inconsistency may arise from the disordered and highly inhomogeneous film structure, composed of both crystalline domains and domain boundaries. While the domain boundaries may significantly affect conductivity, their contribution to the Seebeck coefficient appears to be less pronounced in semicrystalline polymers.⁸⁾ Moreover, it is suggested that the contributions of crystalline domains and boundaries may vary with carrier density, although their impact on thermoelectric properties remains unclear.¹⁰⁾ Therefore, to argue the inconsistency between $S(T)$ and $\sigma(T)$, it is important to simultaneously measure $S(T)$ and $\sigma(T)$ as well as conduct measurements capable of distinguishing between delocalized and localized states, while varying the doping level using a single device.

As a method to distinguish between delocalized and localized states, we focus on magnetoconductivity ($\sigma(B)$) and combine it with a doping-level control technique using electrochemical doping. Electrochemical doping employs an electrolyte as the gate insulator, where the application of a gate voltage drives dopant ions into the polymer film, enabling precise control over the carrier density in the bulk film.^{11–14)} Using this method, we have previously reported a series of S – σ relationships for PBTTT at very high doping levels, beyond the insulator-to-metal transition, using a single device at room temperature. More importantly, by combining this approach with measurements of $\sigma(T)$ and $\sigma(B)$, we clarified the coexistence of two-dimensional weak localization (2D-WL) and variable-range hopping (VRH) processes in a single device at low temperatures. Notably, 2D-WL corresponds to a delocalized transport mechanism, while VRH indicates localized transport, presumably reflecting the inhomogeneous film structure.¹⁴⁾

Although $\sigma(B)$ is the powerful method to distinguish between delocalized and localized states, the comparison among $S(T)$, $\sigma(T)$ and $\sigma(B)$ is lacking.¹⁵⁾ In this study, we designed an electrolyte-gated transistor using the ionic liquid N,N-diethyl-N-(2-methoxyethyl)-N-methylammonium bis(trifluoromethylsulfonyle)imide ([DEME][TFSI]) and developed a measurement system to observe $S(T)$, $\sigma(T)$ and $\sigma(B)$ in PBTTT. By evaluating these properties in a single device, we observed distinct behaviors in both the low- and high-doping regions. In the low-doped region, $S(T)$ exhibits a temperature dependence largely consistent with the VRH model. In contrast, in the highly doped region, we observed that $S(T)$ follows a T -linear relationship at high temperatures, indicating delocalized metallic



behavior, including contributions from 2D-WL. However, at low temperatures, $S(T)$ deviates from this trend and shows an apparent negative intercept at the low-temperature limit. This unusual behavior suggests the coexistence of 2D-WL and VRH processes, as further evidenced by magnetoconductance measurements performed down to 80 mK.

We purchased PBTTC-C14 ($M_w > 50\,000$) from Merck and the ionic liquid [DEME][TFSI] from Kanto Chemical Co., Inc. Figure 1(a) shows the chemical structures of PBTTC and [DEME][TFSI]. For the electrolyte-gated transistor, we thermally evaporated Au/Ni electrodes (30/5 nm) onto a glass substrate (Corning XG, 10 mm $\times$ 10 mm $\times$ 0.5 mm). PBTTC film, with a thickness of 20 nm, was then formed by spin-coating a solution of *o*-dichlorobenzene (7.5 mg ml⁻¹) at 4000 rpm for 120 s. The polymer film was annealed at 190 °C for 30 min. For the electrolyte layer, an ion gel was prepared by mixing the ionic liquid [DEME][TFSI] with the organic polymer P(VDF-HFP) (poly(vinylidene fluoride-co-hexafluoropropylene)) in a 2:1 ratio. The ion gel layer was spin-coated onto a polyethylene naphthalate substrate (Teonex® Q51-16 μ m, Teijin Dupont Films) at 6000 rpm for 120 s, resulting in a film thickness of approximately 100 nm. The ion gel layer was then laminated onto the PBTTC film. The $S(T)$, $\sigma(T)$ and $\sigma(B)$ measurements were performed using a Physical Property Measurement System (PPMS model 6000, Quantum Design), while $\sigma(B)$ measurements at 80 mK were conducted in a dilution refrigerator.

Figures 1(b) and 1(c) show the schematic illustration and optical image of the experimental setup used to measure $S(T)$ and $\sigma(T)$ in electrochemically doped PBTTC films. We adopted a side-gate-type transistor structure with an electrolyte gate insulator.^{13–17} Additionally, an on-chip stripe heater

was integrated to induce a temperature gradient, allowing for measurement of the S .^{8–10,15,18,19} Figure 1(d) shows the transfer characteristics at room temperature, where the drain current (I_D) was recorded with a constant drain voltage ($V_D = -20$ mV) while modulating the gate voltage (V_G). The gate leakage current (I_G) was sufficiently small to be negligible, confirming that the transistor operated as a p-type device. To investigate the effects of doping on the thermoelectric properties, we measured $S(T)$ and $\sigma(T)$ at two doping levels: a low doping level at $V_G = -1.1$ V and a high doping level at $V_G = -2.0$ V. V_G was applied at 300 K, and data were recorded below 180 K, where electrochemical reactions are effectively suppressed due to the freezing of the ion gel.¹³ To determine σ , we measured the current–voltage characteristics at fixed temperatures [Fig. 1(e)], where V_{4pp} represents the voltage drop between the two voltage terminals (V^+ and V^-) in the four-point probe configuration. To measure S , we induced a temperature difference (ΔT) between the source and drain electrodes using Joule heating from the on-chip stripe heater. The temperature of each electrode was determined based on its resistivity. We confirmed that ΔT was proportional to the heater power (P_H), as shown in [Fig. 1(f)]. Figure 1(g) shows the thermoelectromotive force (ΔV) as a function of ΔT at various temperatures, and S was determined from the slope.

Figure 2(a) shows $\sigma(T)$ at a low doping level, $V_G = -1.1$ V, where σ reaches 26 S cm⁻¹ at 180 K. Upon cooling, a significant decrease in σ is observed, dropping to 0.7 S cm⁻¹ at 25 K, which suggests the dominance of a hopping-like localized transport process. To evaluate the localization effect within the framework of the VRH model, we examine $\sigma(T)$ using the Zhabrodskii analysis.¹⁴ In this approach, the logarithmic derivatives of $\sigma(T)$,

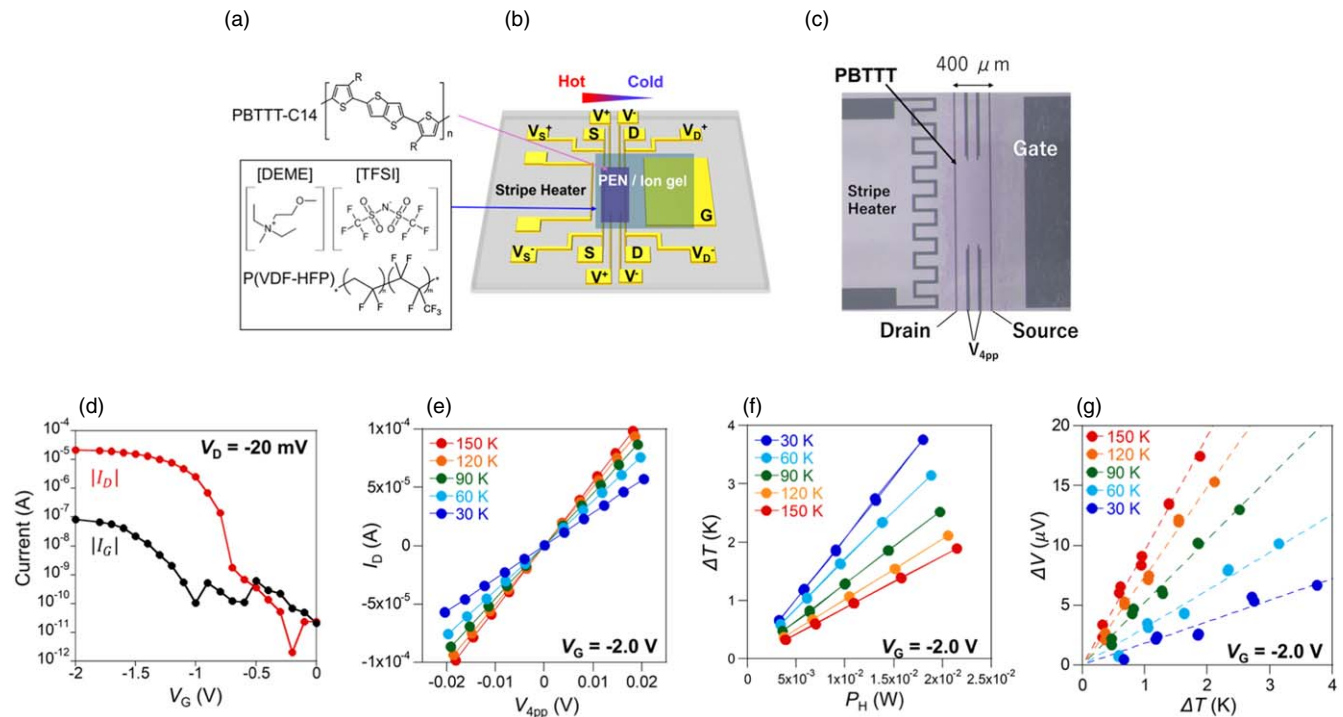


Fig. 1. (a) Chemical structures of PBTTC-C14 and the ion gel composed of [DEME][TFSI] and P(VDF-HFP). (b) Schematic illustration of the thermoelectric measurement system, integrating an electrolyte-gated transistor with an on-chip stripe heater. G, S, and D denote the gate, source, and drain electrodes, respectively. V represents the voltage probe for four-terminal measurements. (c) Optical image of the thermoelectric device. (d) Transfer characteristics measured at room temperature. I_D , I_G , V_G and V_D indicate the drain current, gate leakage current, gate voltage, and drain voltage, respectively. (e) I_D - V_{4pp} characteristics measured at $V_G = -2.0$ V under low-temperature conditions. (f) Heater power (P_H) dependence of the temperature difference (ΔT). (g) Thermoelectromotive force (ΔV) as a function of ΔT in the PBTTC thin film at $V_G = -2.0$ V under low-temperature conditions.

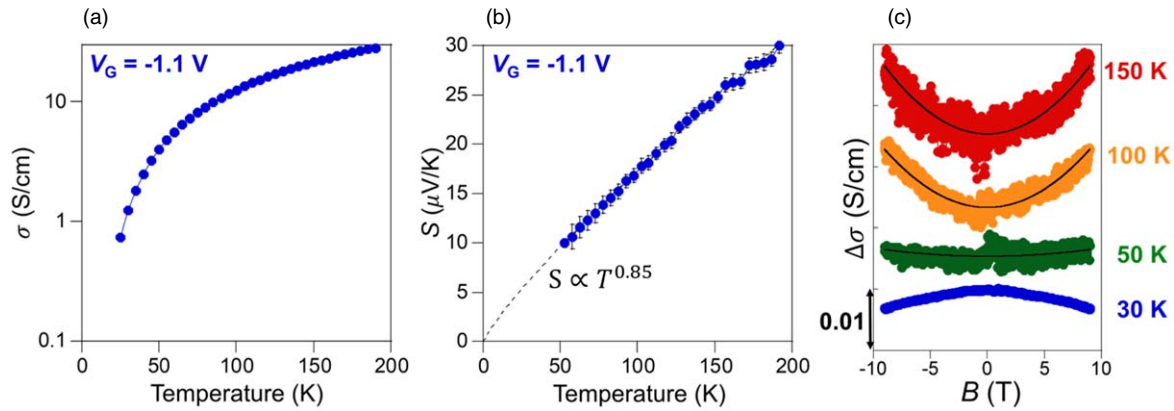


Fig. 2. Temperature dependence of (a) electrical conductivity, (b) Seebeck coefficient and (c) magnetoconductance for PBTTT in the low-doped region at $V_G = -1.1$ V. The black solid curve in (c) represents the Hikami–Larkin–Nagaoka model, with the λ value varying depending on the temperature: $\lambda = 1.89$ nm at 150 K, 1.80 nm at 100 K, and 1.04 nm at 50 K.

$$W = \frac{d(\log \sigma)}{d(\log T)}$$

are plotted on a logarithmic scale as functions of $\log T$. When $\sigma(T)$ follows the Mott VRH model,

$$\sigma(T) = \sigma_0 \exp \left[-\left(\frac{T_0}{T} \right)^a \right]$$

the plot shows a straight line with a slope a . Here, σ_0 is a temperature-independent factor, and T_0 is a characteristic temperature that depends on the localization length and density of states at the Fermi energy. The exponent a indicates the dimensionality of the conduction: $a = 1/4$ for three-dimensional (3D) conduction, $a = 1/3$ for 2D conduction, and $a = 1/2$ for 1D conduction.¹⁴⁾ The Zhabrodsii plots yield straight lines with $a = 0.37$, indicating 2D-Mott VRH (see Supplementary Fig. S1). We also investigated the possibility of 2D-WL, in which $\sigma(T)$ follows a logarithmic temperature dependence, and confirmed that it does not follow this behavior (see Supplementary Fig. S2). Therefore, $\sigma(T)$ at $V_G = -1.1$ V strongly suggests 2D-VRH.

Figure 2(b) shows $S(T)$ at $V_G = -1.1$ V. $S(T)$ exhibits an apparently linear behavior, consistent with Mott's formula for metallic conduction.^{20,21)}

$$S_{\text{Metal}}(T) = -\frac{\pi^2 k_B^2 T}{3e} \left[\frac{d \ln \sigma_E}{dE} \right]_{E_F} \propto T,$$

where e , k_B , σ_E , and E_F are the elementary charge, the Boltzmann constant, the spectral conductivity, and the Fermi level energy, respectively. However, extrapolating the data down to 0 K suggests a positive intercept, which contradicts the physical principle that $S = 0$ at 0 K, as dictated by the third law of thermodynamics (i.e. the entropy approaches a constant minimum at 0 K).²⁰⁾ This inconsistency suggests that, at low temperatures, $S(T)$ may deviate from the metallic behavior and instead follow a sublinear trend, possibly reflecting the dominance of the VRH mechanism.^{1,21,22)} Indeed, a recent theoretical model for disordered systems predicts that $S(T)$ in the VRH regime follows $S_{\text{VRH}}(T) \propto T^{d/(d+1)}$, where d is the dimensionality of system.²²⁾ By fitting our experimental results to this model, we obtained a trend of $S(T) \propto T^{0.85}$ with $d = 5.7$, which lies beyond the range of the model ($d = 2$ or 3). This deviation likely arises from transport processes that do not

strictly adhere to the VRH model, potentially including a weak contribution from delocalized states. These results also highlight the difficulty of making precise arguments based solely on $\sigma(T)$ and $S(T)$.

Finally, to distinguish between delocalized (WL) and localized (VRH) states, we performed magnetoconductivity ($\sigma(B)$) measurements. The magnetic field, B , was applied perpendicular to the substrate, and $\Delta\sigma$ represents the change in σ ($\sigma(B) - \sigma(0)$) upon applying the magnetic field. Figure 2(c) shows $\Delta\sigma$ at $V_G = -1.1$ V, where we observed positive $\Delta\sigma$ at 50, 100, and 150 K, and a negative $\Delta\sigma$ at 30 K. Similar behavior has been reported in studies on highly doped semicrystalline polymers,^{23–25)} with consensus that the positive $\Delta\sigma$ is attributed to the contribution of 2D-WL, as described by the Hikami–Larkin–Nagaoka (HLN) model in two dimensions:²⁶⁾

$$\Delta\sigma_{\text{2D-WL}}(B) \propto -\frac{e^2}{2\pi^2 \hbar} \left[\log \frac{B_\varphi}{B} - \Psi \left(\frac{1}{2} + \frac{B_\varphi}{B} \right) \right]$$

$$B_\varphi = \frac{\hbar}{4e\lambda^2},$$

where $\hbar$ is the reduced Planck constant, $\Psi(x)$ is the digamma function, B_φ is a characteristic magnetic field, and λ is the phase coherence length. The positive $\Delta\sigma(B)$ at 50, 100 and 150 K can be explained by the HLN model with a λ value ranging from 1.0 to 1.9 nm, as shown by the black solid curve in Fig. 2(c). The negative $\Delta\sigma$ at 30 K can be explained by the VRH mechanism (see Supplementary Fig. S3):^{14,27)}

$$\frac{\sigma(B)}{\sigma(0)} \propto \exp(-B^2).$$

These results indicate a crossover of the conduction mechanism from WL to VRH in the low-doping regime as temperature decreases. Moreover, this suggests that simple application of temperature-dependent models for WL or VRH cannot fully explain the observed mechanisms, and the differences between $\sigma(T)$ and $S(T)$ are not necessarily contradictory. Through this analysis, we qualitatively demonstrated a unified discussion of $S(T)$, $\sigma(T)$ and $\sigma(B)$, highlighting the interplay between localized and delocalized transport processes. We emphasize that the contribution of these delocalized transport processes to the total Seebeck

coefficient is subtle but non-negligible in the low-doped PBTTT films.

Figure 3(a) shows $\sigma(T)$ at a high doping level, $V_G = -2.0$ V, where σ exceeds 300 S cm^{-1} at room temperature. This value is comparable to that achieved through electrochemical doping or anion exchange doping in previous reports.^{8,10,13,15,23,24} The Zhabrodsii plots yield straight lines with $a = 0.20$, indicating 3D-Mott VRH (see Supplementary Fig. S4). Additionally, we investigated the possibility of 2D-WL, where $\sigma(T)$ follows a logarithmic temperature dependence, and confirmed that while slight deviations are observed below 50 K, the behavior is well described (see Supplementary Fig. S5). Given the 2D structure of PBTTT, 3D-Mott VRH is unlikely.¹⁴ Therefore, $\sigma(T)$ at $V_G = -2.0$ V strongly suggests 2D-WL.

Figure 3(b) shows $S(T)$ obtained at $V_G = -2.0$ V. Compared to $V_G = -1.1$ V, the Seebeck coefficient is smaller, which is consistent with the increased carrier density. $S(T)$ at $V_G = -2.0$ V appears to follow Mott's formula.²¹ When fitting the low-temperature data with a linear function, we observe a different slope compared to the high-temperature region, suggesting a negative intercept. Notably, similar negative intercepts have been reported in studies using anion exchange doping on PBTTT and,⁹ more recently, observed in electrochemically-doped systems.¹⁵ However, in both cases, the authors did not address the presence of negative intercepts and simply reported the behavior as metallic, implicitly assuming the absence of an intercept and that the data would pass through the origin.

Again, to distinguish between delocalized (WL) and localized (VRH) states, we performed $\sigma(B)$ measurements. Figure 3(c) shows $\Delta\sigma$ at $V_G = -2.0$ V, where we observed positive $\Delta\sigma$ down to 30 K. This behavior can be explained by the HLN model with a λ value ranging from 3.5 to 7.3 nm, as shown by the black solid curve in Fig. 3(c). Therefore, at a high doping level, $V_G = -2.0$ V, we can conclude that the mechanism is 2D-WL, based on $S(T)$, $\sigma(T)$ and $\sigma(B)$. The only remaining issue is the negative intercept observed in $S(T)$. It is important to emphasize that such negative intercepts cannot be explained by either WL or VRH, both of which inherently predict intercept-free behavior. Given that the mechanism at a high doping level is concluded to be 2D-WL, this unusual temperature dependence may suggest a minor contribution to the total $S(T)$ data from coexisting

VRH, owing to the small S values in the highly doped 2D-WL regime. To further explore the origin of this behavior, we conducted low-temperature magnetoconductance measurements, which provide clearer insights into the coexisting VRH states.

Figures 4(a) and 4(b) show $\sigma(B)$ at the maximum doping level for 4.2 K and 80 mK, respectively. At 4.2 K, we observed a positive $\Delta\sigma$ component at low magnetic fields and a negative $\Delta\sigma$ component at high magnetic fields. The negative $\Delta\sigma$ component observed at high magnetic fields is attributed to the contribution of the VRH mechanism (see Supplementary Fig. S6).²⁷ In contrast, the $\Delta\sigma$ in the low magnetic field region can be described by the combined contributions of the VRH mechanism and HLN model (the 2D-WL mechanism with a λ value of 15.4 nm, see Supplementary Fig. S7). This combined behavior is represented by the black solid curve in Fig. 4(a). These results clearly demonstrate that, even at the maximum doping level, the contribution of VRH cannot be neglected and becomes more significant at lower temperatures, indicating a crossover from 2D-WL to VRH and the coexistence of both mechanisms. To the best of our knowledge, this coexistence of 2D-WL and VRH is a phenomenon unique to semicrystalline polymers, likely arising from film inhomogeneity.

Specifically, in crystalline domains, the tightly packed and stable structure due to molecular stacking undergoes minimal structural changes upon doping. These changes are limited to lattice expansion in the lamellar direction and slight compression of π - π stacking distances.¹² In contrast, domain boundaries, with their more amorphous-like structure and increased molecular spacing, are more susceptible to structural changes upon doping. In our previous study, using DFT calculations, we performed structural optimization of the dimer unit in a single PBTTT chain in the neutral and cationic states.¹² The results revealed that the planarity of the molecules improves significantly in the cationic state, facilitating metallic conduction. Therefore, we propose that in domain boundaries, doping induces not only an increase in carrier density but also structural changes, leading to enhanced uniformity in terms of dihedral angles across the film. This structural improvement may further contribute to the observed crossover from 2D-WL in the highly doped samples to VRH in the low-doped samples, with delocalized states occurring predominantly in crystalline domains and localized

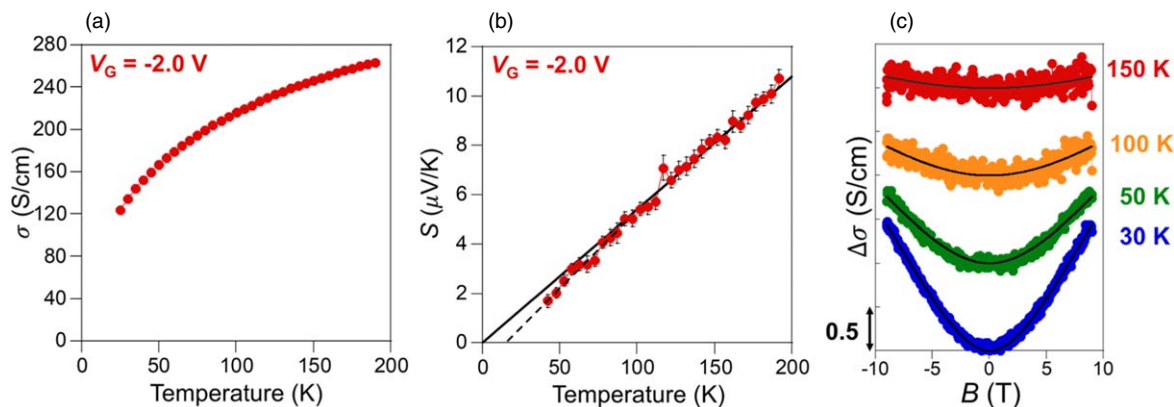


Fig. 3. Temperature dependence of (a) electrical conductivity, (b) Seebeck coefficient and (c) magnetoconductance for PBTTT in the high-doped region at $V_G = -2.0$ V. The black solid line in (b) represents the Mott equation, while the dashed line serves as an eye guide for the linear function below 80 K. The black solid curve in (c) represents the Hikami-Larkin-Nagaoka model, with the λ value varying depending on the temperature: $\lambda = 3.51$ nm at 150 K, 4.56 nm at 100 K, 5.84 nm at 50 K, and 7.28 nm at 30 K.

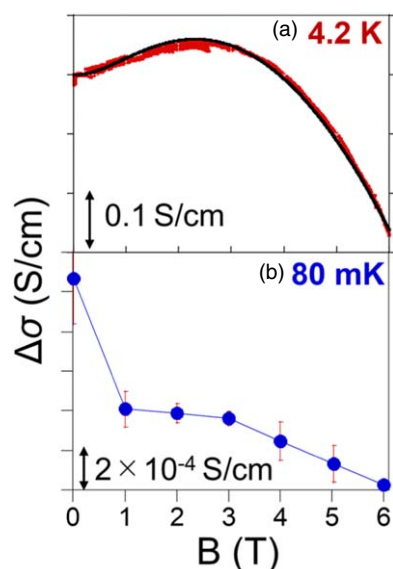


Fig. 4. Temperature dependence of the magnetoconductivity of a highly doped PBTTT film at (a) 4.2 K and (b) 80 mK. The black solid curve in (a) represents the combined contributions of the VRH mechanism (see Supplementary Fig. S6) and HLN model (the 2D-WL mechanism with a λ value of 15.4 nm, see Supplementary Fig. S7).

states persisting at domain boundaries. As the temperature decreases, thermal excitation from trap states is suppressed, reducing the carrier density. Consequently, the system transitions from the cationic to the neutral state. This change likely alters the molecular dihedral angles in the boundary regions, leading to a loss of conductivity and a transition from 2D-WL to VRH.

We should address the differences between low-doping-level and high-doping-level samples. As shown in Fig. 2(c), negative $\Delta\sigma$ is observed in the low-doping-level sample, indicating the presence of VRH. However, unlike the high-doping-level samples at 4.2 K, no coexistence with 2D-WL is observed in the low-doping regime. Considering that the contribution of VRH becomes stronger at lower temperatures, this clearly demonstrates that VRH becomes the dominant transport mechanism in the low-temperature regime of low-doping samples. These findings highlight a key distinction: the coexistence of 2D-WL and VRH is a unique feature of highly doped samples, where delocalized and localized states coexist due to film inhomogeneity and increased carrier density. This coexistence does not occur in low-doped samples, emphasizing that the interplay between these mechanisms is strongly dependent on both temperature and doping level.

At 80 mK, $\sigma(B)$ was measured only after ensuring that the temperature had fully stabilized, thereby eliminating any potential effects of temperature rise caused by the magnetic field sweep. At low magnetic fields (up to 1 T), a steep negative magnetoconductance was observed. Although the exact mechanism behind this behavior remains an open question, its absence at 4.2 K in Fig. 4(a) suggests that it may originate from shallower carrier trapping. This could be attributed to minor structural disorder or Coulomb trapping by dopant ions within the crystalline domains, in contrast to the VRH process occurring at domain boundaries, which dominates in the high magnetic field region at 4.2 K. In the high magnetic field region ($B > 3$ T), the gradual negative

magnetoconductance was explained by the VRH mechanism, consistent with the results at 4.2 K (see Supplementary Fig. S8). The nearly flat magnetic field dependence observed in the intermediate field range ($1 \text{ T} < B < 3 \text{ T}$) indicates a balance between the negative VRH component and a positive contribution attributed to the 2D-WL process within the crystalline domains. This 2D-WL behavior has previously been reported in electrochemically doped PBTTT at temperatures as low as 1.9 K.¹⁴⁾ These findings suggest that the coexistence of 2D-WL and VRH processes persists even at 80 mK. Understanding this coexistence is essential for elucidating the origin of the anomalous Seebeck coefficient behavior in this system, emphasizing the need for further theoretical investigations to clarify the underlying mechanisms.

In summary, we developed a system that enables $S(T)$, $\sigma(T)$ and $\sigma(B)$ measurements under electrochemical doping control. In the low-doped region, we observed that $S(T)$ is predominantly influenced by the VRH transport process. In the highly doped region, $S(T)$ is reasonably described by Mott's formula, characteristic of metals at high temperatures above 80 K. On the other hand, at lower temperatures, $S(T)$ exhibits an unusual downward deviation from Mott's formula, which cannot be explained by conventional transport models. We suggest that $S(T)$ is affected by the interplay between delocalized and localized states, the coexistence of which is confirmed by magnetoconductance measurements down to 80 mK.

Supplementary data

See the supplementary Figures S1 to S8.

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- 1) O. Bubnova and X. Crispin, *Energy Environ. Sci.* **5**, 9345 (2012).
- 2) S. N. Patel, A. M. Glaudell, K. A. Peterson, E. M. Thomas, K. A. O'Hara, E. Lim, and M. L. Chabiny, *Sci. Adv.* **3**, e1700434 (2017).
- 3) A. M. Glaudell, J. E. Cochran, S. N. Patel, and M. L. Chabiny, *Adv. Energy Mater.* **5**, 1401072 (2015).
- 4) B. Russ, A. Glaudell, J. J. Urban, M. L. Chabiny, and R. A. Segalman, *Nat. Rev. Mater.* **1**, 16050 (2016).
- 5) H. Abdalla, G. Zuo, and M. Kemerink, *Phys. Rev. B* **96**, 241202(R) (2017).
- 6) S. D. Kang and G. J. Snyder, *Nat. Mater.* **16**, 252 (2017).
- 7) S. A. Gregory, R. Hanus, A. Atassi, J. M. Rinehart, J. P. Wooding, A. K. Menon, M. D. Losego, G. J. Snyder, and S. K. Yee, *Nat. Mater.* **20**, 1414 (2021).
- 8) S. Watanabe, M. Ohno, Y. Yamashita, T. Terashige, H. Okamoto, and J. Takeya, *Phys. Rev. B* **100**, 241201 (2019).
- 9) Y. Huang et al., *Appl. Phys. Lett.* **119**, 111903 (2021).
- 10) C. Chen et al., *Adv. Energy Mater.* **13**, 2202797 (2023).
- 11) H. Tanaka, S. Nishino, H. Ito, and S.-I. Kuroda, *Appl. Phys. Lett.* **107**, 243302 (2015).
- 12) T. Harada, H. Ito, Y. Ando, S. Watanabe, H. Tanaka, and S.-I. Kuroda, *Appl. Phys. Express* **8**, 021601 (2015).
- 13) H. Tanaka, K. Kanahashi, N. Takekoshi, H. Mada, H. Ito, Y. Shimoi, H. Ohta, and T. Takenobu, *Sci. Adv.* **6**, eaay8065 (2020).
- 14) H. Ito, H. Mada, K. Watanabe, H. Tanaka, and T. Takenobu, *Commun. Phys.* **4**, 8 (2021).

- 15) D. H. L. Tjhe et al., *Nat. Mater.* **23**, 1712 (2024).
- 16) K. Kanahashi, Y.-Y. Noh, W.-T. Park, H. Yang, H. Ohta, H. Tanaka, and T. Takenobu, *Phys. Rev. Res.* **2**, 043330 (2020).
- 17) S. Ito, K. Kanahashi, H. Ohta, H. Ito, T. Takenobu, and H. Tanaka, *Appl. Phys. Lett.* **119**, 183304 (2021).
- 18) D. Venkateshvaran et al., *Nature* **515**, 384 (2014).
- 19) D. Venkateshvaran, A. J. Kronemeijer, J. Moriarty, D. Emin, and H. Sirringhaus, *APL Mater.* **2**, 032102 (2014).
- 20) A. Sommerfeld, *Thermodynamics and Statistical Mechanics* (Academic, New York, 1964).
- 21) N. M. Mott and E. A. Davis, *Electron Processes in Non-crystalline Materials* (Clarendon, Oxford, 1979) 2nd ed. p. 55.
- 22) T. Yamamoto, M. Ogata, and H. Fukuyama, *J. Phys. Soc. Jpn.* **91**, 044704 (2022).
- 23) Y. Yamashita, J. Tsurumi, M. Ohno, R. Fujimoto, S. Kumagai, T. Kurosawa, T. Okamoto, J. Takeya, and S. Watanabe, *Nature* **572**, 634 (2019).
- 24) S. Zanettini, J. F. Dayen, C. Etrillard, N. Leclerc, M. Venkata Kamalakar, and B. Doudin, *Appl. Phys. Lett.* **106**, 063303 (2015).
- 25) K. Kang et al., *Nat. Mater.* **15**, 896 (2016).
- 26) S. Hikami, A. I. Larkin, and Y. Nagaoka, *Prog. Theor. Phys.* **63**, 707 (1980).
- 27) B. I. Shklovskii and A. L. Efros, *Electronic Properties of Doped Semiconductors* (Springer, New York, 1984).